

Exploratory Data Analysis Report

Retail Sales Dataset Analysis

Prepared as part of Data Science Internship – Task 1

Dataset Source: Kaggle – Retail Sales Dataset

<https://www.kaggle.com/datasets/mohammadtalib786/retail-sales-dataset>

Date: July 2026

1. Introduction

This report presents an Exploratory Data Analysis (EDA) conducted on the Retail Sales Dataset, sourced from Kaggle. The dataset contains transactional records from a retail business, capturing customer demographics, product categories, purchase quantities, and sales amounts. The objective of this analysis is to understand sales patterns, customer behavior, and relationships between key variables in order to derive actionable business insights.

Dataset Overview:

- Source: Kaggle – Retail Sales Dataset
- Type: Transactional retail sales data
- Key Fields: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, Total Amount
- Product Categories: Beauty, Clothing, Electronics

2. Data Cleaning & Preprocessing

Before analysis, the following preprocessing steps were carried out to ensure data quality and consistency:

- Checked and removed duplicate transaction records.
- Verified missing values across all columns using null-value counts.
- Converted the 'Date' column from text to a proper datetime format to enable time-based analysis.
- Created a derived 'Month' field to support monthly trend analysis.
- Created a derived 'Age Group' field (binned into ranges) to support demographic analysis.
- Used boxplot visualization to inspect the 'Total Amount' column for outliers.

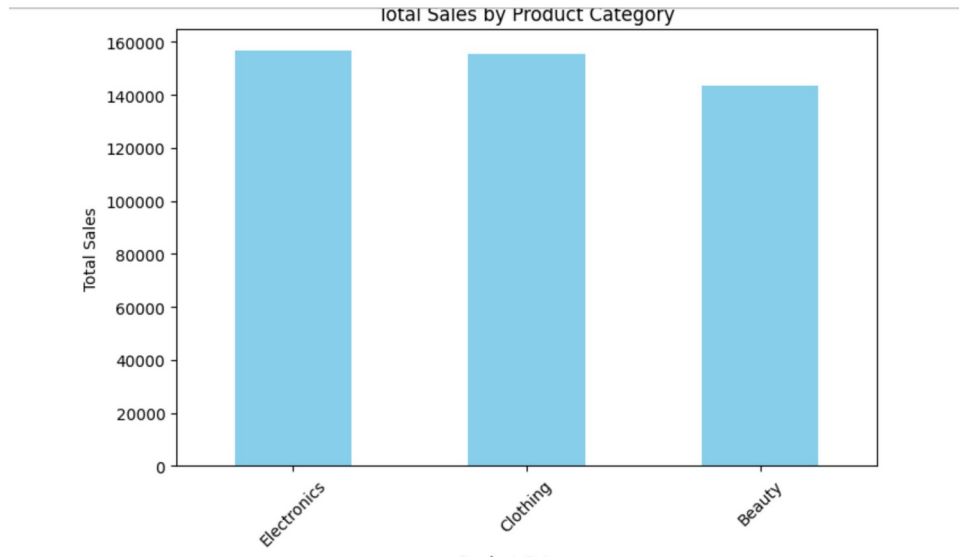
The boxplot analysis (see Section 4) showed that while there is a wide spread in transaction values, extreme values remain within the natural range of the business (high-value Electronics purchases), and were therefore retained as valid data rather than removed as errors.

3. Key Insights

- Sales are fairly evenly distributed across the three product categories, with Electronics generating the highest total sales (~157,000), followed closely by Clothing (~156,000) and Beauty (~144,000).
- Customer spending is almost evenly split by gender, with Female customers contributing 51.1% of total sales and Male customers 48.9% — indicating no strong gender-based skew in purchasing behavior.
- Customer age is fairly uniformly distributed between 20 and 64 years, with no single age group dominating the customer base, suggesting the store appeals to a broad demographic.
- Monthly sales show clear fluctuation: May 2023 recorded the highest sales (~53,000), while September 2023 recorded the lowest (~24,000). The sharp drop in Jan 2024 reflects an incomplete final month in the dataset rather than an actual decline.
- Price per Unit shows a strong positive correlation with Total Amount (0.85), meaning higher-priced items are the primary driver of transaction value.
- Quantity shows a moderate positive correlation with Total Amount (0.37), while Age shows almost no correlation with spending (-0.06), indicating age is not a strong predictor of purchase value.
- The scatter plot confirms that Electronics purchases at higher price points (around 500 per unit) generate the highest individual transaction values, reaching up to 2,000.

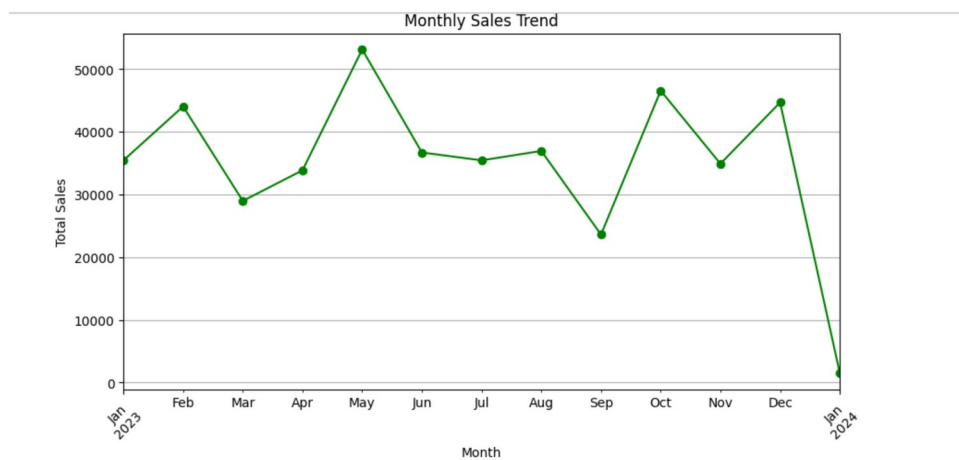
4. Data Visualizations

4.1 Total Sales by Product Category



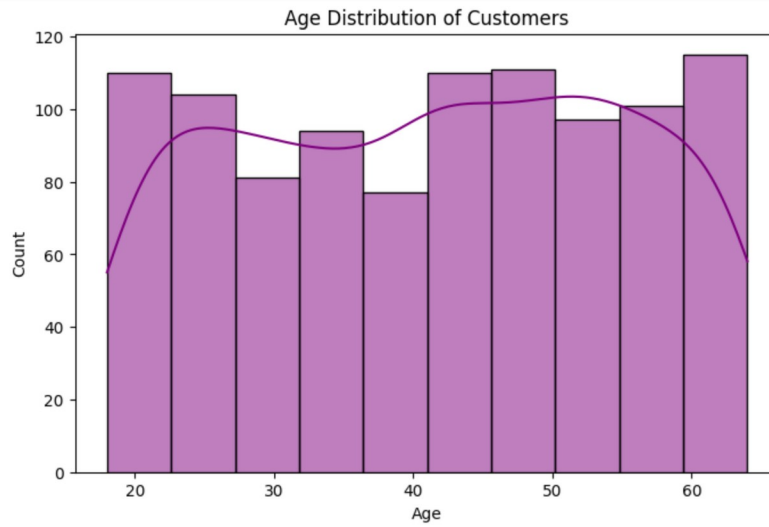
Electronics leads in total sales, closely followed by Clothing, with Beauty slightly lower.

4.2 Monthly Sales Trend



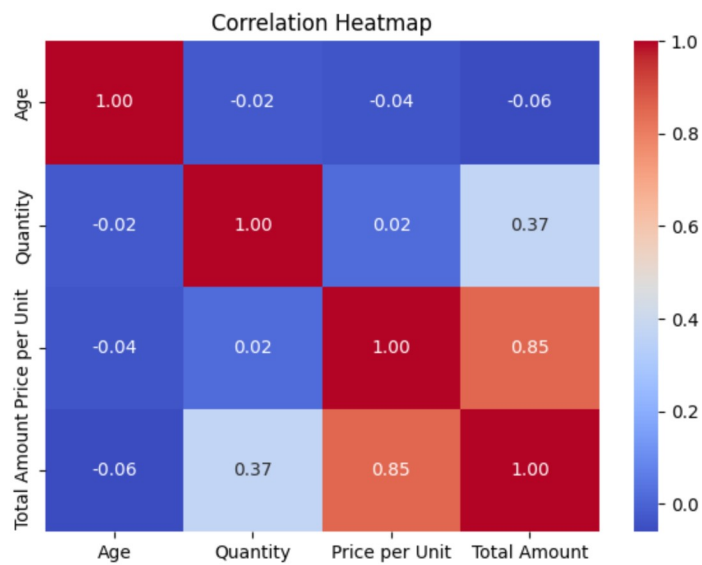
Sales fluctuate month to month, peaking in May 2023 and dipping in September 2023.

4.3 Age Distribution of Customers



Customer ages are broadly and fairly uniformly distributed between 20 and 64 years.

4.4 Correlation Heatmap



Price per Unit has the strongest correlation with Total Amount (0.85).

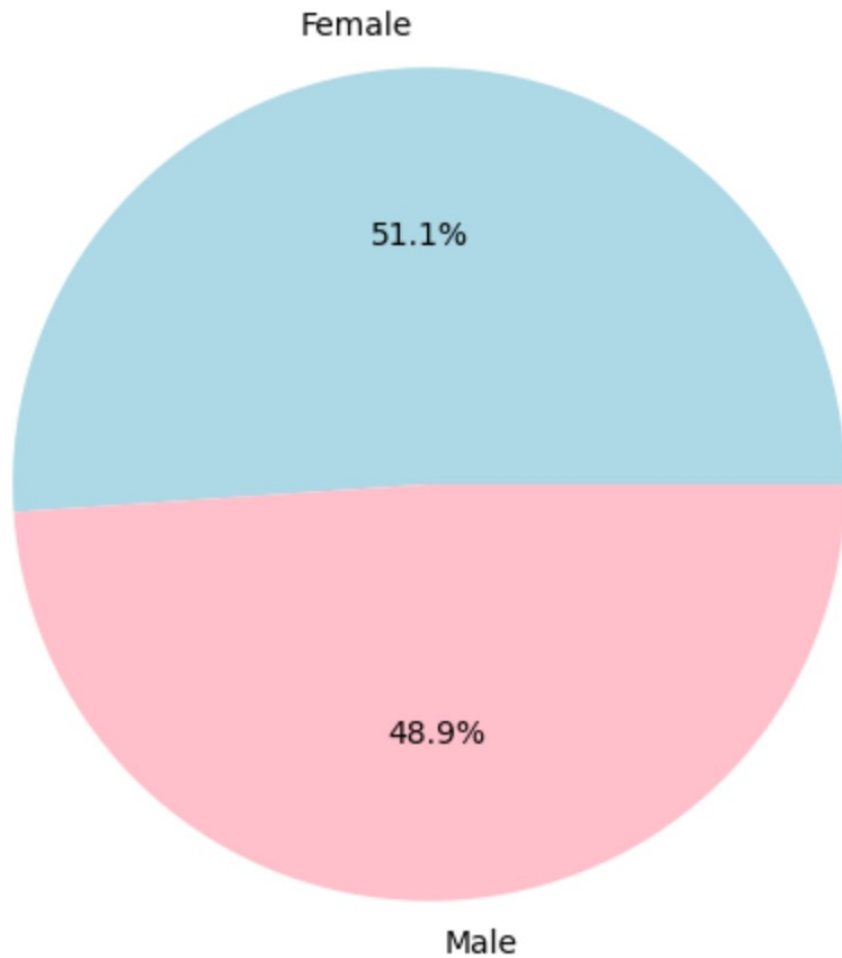
4.5 Price per Unit vs Total Amount (Scatter Plot)



Higher-priced Electronics items drive the highest total transaction amounts.

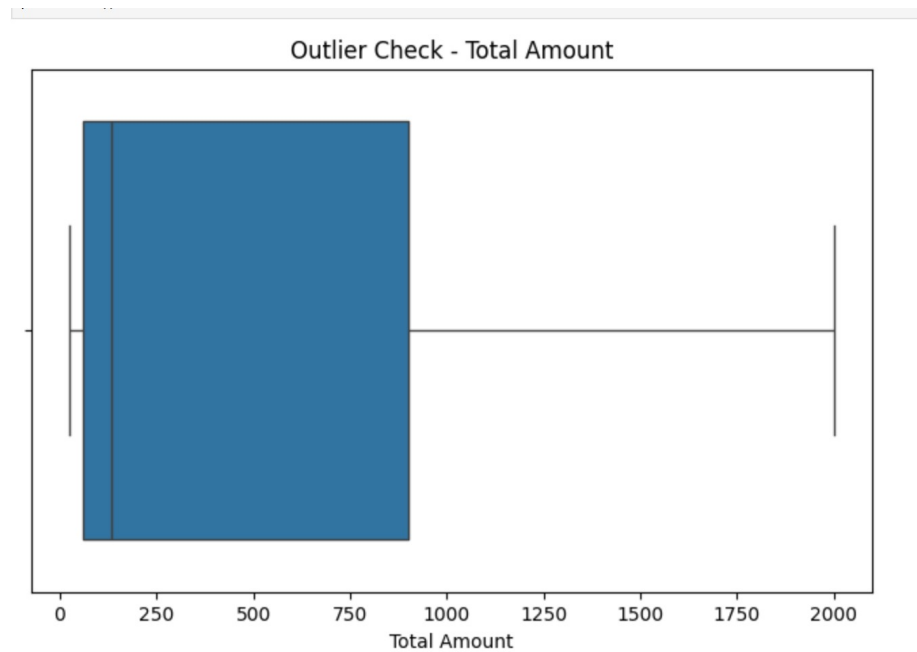
4.6 Sales Distribution by Gender

Sales Distribution by Gender



Sales are nearly evenly split between Female (51.1%) and Male (48.9%) customers.

4.7 Outlier Check – Total Amount



Boxplot shows a wide but reasonable spread in transaction values, with no invalid extreme outliers.

5. Business Recommendations

- Invest in Electronics inventory and marketing, as it is the top revenue-generating category and shows the highest per-transaction value potential.
- Since spending is evenly split by gender and age, marketing campaigns should remain broad and inclusive rather than narrowly targeted at one demographic.
- Launch promotional campaigns or discounts in low-sales months (e.g., September) to smooth out seasonal dips and stabilize monthly revenue.
- Focus upselling efforts on increasing average Price per Unit or bundling higher-value items, since price (not quantity or age) is the strongest driver of transaction value.
- Continue monitoring high-value Electronics transactions, as they represent important but variable revenue events worth tracking for demand forecasting.

6. Conclusion

The exploratory analysis of the Retail Sales Dataset reveals a balanced business across product categories and customer demographics, with product price being the strongest driver of sales value. These insights provide a foundation for targeted inventory planning, seasonal promotions, and pricing strategy to support continued revenue growth.